

Uncertainty Quantification for a Turbulent Round Jet Using Multifidelity Karhunen-Loève Expansions

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Understanding the behavior of turbulent jets under variable environment and uncertain conditions is critical for predicting and mitigating aircraft jet noise. However, uncertainty quantification (UQ) for jet noise, which requires repeated expensive eddy-resolving simulations, is often computationally prohibitive. We thus build surrogate models, in particular Karhunen-Loève expansions (KLEs) for field quantities of interest in three-dimensional turbulent round jets. We build them in a multifidelity manner by combining simulation data from high-fidelity enhanced delayed detached-eddy simulation (EDDES) and low-fidelity Reynolds-averaged Navier-Stokes (RANS), generated under uncertain nozzle exit stagnation pressure and inlet eddy viscosity ratio. Furthermore, we form the KLEs in conjunction with polynomial chaos expansions in order to explicitly associate their randomness to each physical source of uncertainty, and so justifying the combining procedure in the multifidelity construct. We illustrate advantages of the new multifidelity KLE against single-fidelity KLEs, with the former achieving more accurate predictions at locations away from existing high-fidelity training data. With the KLE surrogate, we conduct UQ inexpensively.

I. Nomenclature

$\sim \mathcal{U}(a, b)$	= uniform distribution between a and b
A	= regression matrix
$b_{k,\beta}$	= polynomial chaos coefficients
c	= coefficient vector in regression problem
D	= nozzle diameter
$\mathcal{I}$	= index set
$K(\cdot, \cdot), K$	= covariance kernel and matrix
k_t	= truncation index
M_f	= flight Mach number
M_j	= jet Mach number
N, N_1, N_2	= number of samples
n_g	= number of grid points
n_s	= stochastic degrees of freedom
n_t	= number of expansion terms
p	= polynomial degree
p_0	= stagnation pressure, Pa
p_0^*	= nominal value of stagnation pressure, Pa

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Q	=	eigenvector matrix
$q_k(\cdot), q_k$	=	k th eigenfunction and eigenvector
Re_D	=	Reynolds number based on the nozzle diameter
w_j, W	=	weight for the j th point, weight matrix
x/D	=	streamwise coordinates in jet flow
$\mathbf{x}$	=	spatial coordinate
y	=	random field quantity
$\tilde{y}$	=	approximation to y
$\beta^{(j)}$	=	multi-index for $j = 1, \dots, n_t$
Δ	=	additive correction or discrepancy term in multifidelity model
Γ	=	weight matrix for Tikhonov regularization
ζ_k	=	random variables in Karhunen-Loève expansion
Λ	=	eigenvalue matrix
λ_k	=	k th eigenvalue
μ	=	mean of random field
$\tilde{\mu}$	=	sample mean of random field
$\tilde{\nu}$	=	modified eddy viscosity
ν	=	kinematic viscosity
ξ_j	=	reference random variables in polynomial chaos expansion
τ	=	regularization parameter
Ψ_{β^j}	=	multivariate orthonormal polynomial basis functions
$\psi_{\beta_i^{(j)}}$	=	univariate orthonormal polynomial basis functions
ω	=	sample space random event

II. Introduction

Turbulent mixing of jets exiting the aircraft engine nozzle is a significant contributor to the overall aircraft noise. While jet noise has been drastically reduced since the advent of high-bypass ratio engines, it remains a major source of aircraft noise at takeoff when the engines are operating at maximum thrust. Furthermore, as various components of the airframe noise are being reduced by different emerging technologies, the jet-flap interaction is perceived as a prominent noise contributor. It is therefore essential to understand and accurately predict the flow of turbulent jets and the associated noise generation mechanisms in the quest to suppress jet-related aircraft noise.

For example, it has been shown that the difference between the flight and the jet Mach numbers strongly modifies the jet shear layer that is responsible for noise generation, and a static jet and a jet in flight can produce very different far-field noise signatures [1]. Numerous experimental studies in turbulent jets also demonstrated that the turbulence intensity and stagnation pressure at nozzle exit plane can greatly modify the turbulent flow physics in the jet shear layer [2–4]. With variable environment and uncertain conditions often encountered in real life, an important step then is to understand and quantify these sources of uncertainty and assess how they impact the physical behavior of turbulent jets. We will investigate these effects leveraging numerical simulations.

A statistical characterization on the influence of uncertainty sources is often sought through uncertainty quantification (UQ), which typically entails the task of “propagating” the uncertainty from model inputs to outputs. For investigation of jets, Mishra *et al.* [5, 6] perturbed the Reynolds stress eigenvalues and eigenvectors to estimate the uncertainty due to closure assumptions in several Reynolds-averaged Navier-Stokes (RANS) jet simulations, but that study did not yet include other uncertainty sources, and constructed “uncertainty bounds” with only 5 prescribed perturbation cases. A complete distribution description of uncertainty can be captured with methods such as Monte Carlo (MC), where one generates random samples of the input variables from its probability distribution, and then performs a flow solve for each sample. MC is non-intrusive, simple to implement, and embarrassingly parallel; but it quickly becomes computationally prohibitive if each simulation is already expensive, often the case for turbulent flows.

One strategy to alleviate this high computational cost is to construct a surrogate model to replace the expensive original model for use in the UQ computations, where the surrogate aims to provide fast and accurate approximations to the input-output relationship directly for our model quantities of interest (QoIs). In this paper, we are specifically interested in building surrogates for *field* QoIs (e.g., QoIs that are functions of space and/or time). When there is randomness in the model, the field QoIs are characterized by random fields (RFs) (or more generally, stochastic processes), which is in contrast to random variables/vectors often used for representing finite-dimensional QoIs.

On this front, Granados-Ortiz *et al.* [2] illustrated the use of spatially-dependent polynomial chaos expansions (PCEs) and kriging (Gaussian process regression) as surrogate models for RFs in three-dimensional RANS simulations of jet flows experiencing variability of nozzle inlet stagnation pressure and viscosity ratio. In our work, we will explore another surrogate choice for RFs in the form of Karhunen-Loève expansions (KLEs) [7]. KLEs are expansions for general stochastic processes based on the spectral expansion of their covariance functions. They offer attractive properties with their proven spectral convergence rates, intuitive way to truncate, ability to represent RFs beyond Gaussian processes, and doing so in a parametric fashion. Furthermore, we form the KLEs in conjunction with PCEs, in a manner that allows us to explicitly associate their randomness to each physical source of uncertainty [7, 8]; this idea will be crucial for establishing our multifidelity KLE formulation introduced later.

The aforementioned jet studies all involved RANS computations of near-field hydrodynamic quantities. However, for jet noise, high-fidelity eddy-resolving methods such as large eddy simulation (LES) or hybrid RANS/LES simulations are necessary to resolve the noise generating turbulent fluctuations in the jet shear layer [9]. In this work, we use the enhanced delayed detached-eddy simulation (EDDES) with a shear-layer adaptive sub-grid scale model that has demonstrated excellent performance in predicting separated turbulent flows [10, 11]. While these high-fidelity numerical simulations for turbulent jets in recent years have been shown to produce predictions in very good agreement with experiments [1, 12], they are typically too expensive to generate sample sizes large enough for a meaningful UQ analysis, either directly via MC or for building surrogate models. At the same time, a greater number of simulations can be obtained if one opts instead for a lower fidelity model, such as with RANS or a coarser grid. This then motivates the exploration for jointly using multiple models of varying fidelity—a *multifidelity* approach—in order to take advantage of their accuracy-versus-cost tradeoffs and achieve an overall efficient computation.

We take a multifidelity approach to conduct UQ [13, 14]. Our notion of model fidelity is quite general, for example while the high-fidelity (HF) model may be LES, the low-fidelity (LF) model can result from reduced physics, reduced order, coarsened mesh, or direct functional approximations. In any case, the LF models must provide some useful information about the HF models. Together with a model management strategy, the computational work can then be distributed among the different models and to establish accuracy and convergence of the results [13]. The most prominent group of multifidelity UQ work is perhaps in MC sampling. These techniques leverage the variance decay among grid discretization levels (multilevel MC [15]), control variates (multifidelity MC [16]), or a combination of both (multilevel multifidelity MC [17–19]). We do not directly employ these MC techniques, but instead aim to construct a surrogate model, similar to those previously investigated for polynomial chaos and stochastic collocation [20–22]. Here, a discrepancy model is constructed and can be used as a correction term to the LF model predictions. It can be in additive or multiplicative forms, and adaptively updated as new simulations become available [13]. However, we currently do not build such an adaptation model management strategy in this paper, but instead take the first step to construct multifidelity KLE surrogate models given fixed training data of turbulent round jet simulations.

The key novelty and contributions of our paper are summarized as follows.

- We will move beyond RANS, and perform EDDES simulations of a turbulent round jet that will accurately resolve the noise generating turbulent fluctuations in the jet shear layer.
- We will conduct UQ of field QoIs through KLE surrogates constructed under a multifidelity framework, leveraging a combination of HF EDDES runs together with LF RANS simulations.

This paper is organized as follows. Section III.A describes the modelling and setup for obtaining simulations of the turbulent jet. Sections III.B to III.D describe the surrogate model we will use for UQ, namely the KLE and its construction in a multifidelity setting. Section IV presents the multifidelity results for the single-stream round jet simulations. Lastly, conclusions and future work can be found in Section V.

III. Methods

A. Turbulent Jet Simulation Using SU2

We carry out all our jet simulations using SU2 [23], an open-source software suite specifically developed for solving problems governed by partial differential equations (PDEs) and PDE-constrained optimization. SU2 was created with the aerodynamic shape optimization problems in mind, and so it is centered around a RANS solver capable of simulating compressible, turbulent flows commonly found in problems in aerospace engineering. The governing equations are spatially discretized using the finite volume method on unstructured meshes. A number of convective fluxes discretization schemes have been implemented, such as the Jameson-Schmidt-Turkel (JST) scheme and the upwind Roe scheme. The turbulence can be either modeled by the Spalart-Allmaras (S-A) model or the Menter Shear

Stress Transport (SST) Model. For unsteady flows, a second-order dual time-stepping method can be used to obtain time-accurate solutions.

For scale-resolving capabilities, the EDDDES based on the S-A model was implemented in SU2 by Molina [24] and has been demonstrated to successfully predict separated flows [10, 11]. To mitigate the “grey area” problem characterized by slow transition from RANS to LES mode in the shear-layer, a shear-layer adaptive (SLA) sub-grid scale model [25] was implemented. Additionally, to limit the numerical dissipation in LES part of the EDDDES model, the inviscid flux is computed using the so-called simple low dissipation advection upstream (SLAU2) [26]. For this work, the EDDDES-SA solver will be used as the HF model for the turbulent jet simulation while the RANS-SA solver will be used as its LF counterpart.

B. Karhunen-Loève Expansions of Random Fields

We aim to build KLEs as surrogate models for field QoIs (e.g., quantities that are functions of space and/or time). Consider a QoI $y(\mathbf{x}, \omega)$ portrayed as a RF that is dependent on the spatial coordinates $\mathbf{x} \in \mathcal{X}$ and random event $\omega \in \Omega$, where $\mathcal{X}$ is the spatial domain and Ω is the sample space representing all sources of uncertainty. We further write

$$y(\mathbf{x}, \omega) = \mu(\mathbf{x}) + y_0(\mathbf{x}, \omega) \quad (1)$$

where $\mu(\mathbf{x}) = \mathbb{E}[y(\mathbf{x}, \omega)]$ is the RF mean, and $y_0(\mathbf{x}, \omega)$ is the remainder RF that itself has zero mean. The KLE focuses on the centered (zero mean) RF $y_0(\mathbf{x}, \omega)$. When the RF is square integrable and continuous in the mean square sense, it can be expressed via a spectral decomposition known as the KLE:

$$y_0(\mathbf{x}, \omega) = \sum_{k=1}^{\infty} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega), \quad (2)$$

where $\zeta_k(\omega)$ are mutually uncorrelated random variables with zero mean and unit variance. The scalars $\lambda_1 \geq \lambda_2 \geq \dots \rightarrow 0$ and functions $q_k(\mathbf{x})$ are respectively the eigenvalues and orthonormal eigenfunctions to the homogeneous Fredholm equation of the second kind:

$$\int_{\mathcal{X}} K(\mathbf{x}_i, \mathbf{x}_j) q_k(\mathbf{x}_j) d\mathbf{x}_j = \lambda_k q_k(\mathbf{x}_i), \quad k = 1, 2, \dots, \quad (3)$$

where

$$K(\mathbf{x}_i, \mathbf{x}_j) = \mathbb{E}[y_0(\mathbf{x}_i, \omega) y_0(\mathbf{x}_j, \omega)] \quad (4)$$

is the covariance function of $y_0(\mathbf{x}, \omega)$. The KLE can be shown to be optimal, where a finite truncation yields the minimum attainable mean square error.

Following [27], we build our KLEs in a data-driven manner using a training dataset $\{y(\mathbf{x}_j, \omega^{(n)})\}$, where $n = 1, \dots, N$ indexes the N flow solve simulations obtained under different realizations of the uncertain inputs, and $j = 1, \dots, n_g$ indexes the n_g grid points in each simulation (currently we assume n_g fixed for all N runs for simplicity). The first step is to extract the sample mean $\tilde{\mu}(\mathbf{x}_j)$ (we use the tilde notation $\tilde{\cdot}$ to denote approximation) and use it to center the dataset and obtain $\{y_0(\mathbf{x}_j, \omega^{(n)})\}$. To accommodate non-uniform grids, we also introduce weights $w_j > 0$ that reflect the local cell area associated with each node. The discretized form of Eq. (3) is then

$$\int_{\mathcal{X}} K(\mathbf{x}_i, \mathbf{x}_j) q_k(\mathbf{x}_j) d\mathbf{x}_j = \lambda_k q_k(\mathbf{x}_i) \quad \longrightarrow \quad \sum_{j=1}^{n_g} w_j K_{ij} q_{k,j} = \lambda_k q_{k,i}, \quad i = 1, \dots, n_g \quad (5)$$

where we adopted notations $K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$ and $q_{k,j} = q_k(\mathbf{x}_j)$, and

$$K_{ij} = \frac{1}{N-1} \sum_{n=1}^N y_0(\mathbf{x}_i, \omega^{(n)}) y_0(\mathbf{x}_j, \omega^{(n)}), \quad i, j = 1, \dots, n_g. \quad (6)$$

Equation (5) in matrix form is then

$$(KW)Q = Q\Lambda. \quad (7)$$

where $K \in \mathbb{R}^{(n_g \times n_g)}$, $W \in \mathbb{R}^{(n_g \times n_g)}$ is the diagonal quadrature weight matrix with $W_{jj} = w_j$, and Q and Λ are the right eigenvector matrix and the eigenvalue matrix, respectively. If $n_g \gg N$, Eq. (7) can be transformed into a singular value decomposition problem for memory savings (see [27]).

The solution to Eq. (7) produces $\min(n_g, N - 1)$ non-zero modes, but in practice we can truncate the KLE further with a smaller (often much smaller) k_t terms while retaining good accuracy:

$$y_0(\mathbf{x}, \omega) \approx \tilde{y}_0(\mathbf{x}, \omega) = \sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega). \quad (8)$$

k_t is typically selected upon assessing the the rate of eigenvalue decay, for example such that λ_k has decayed to some fraction (e.g., 10%) of the largest eigenvalue λ_1 . For each mode k , we can compute N sample realizations of $\zeta_k(\omega^{(n)})$ from the N simulations through

$$\zeta_k(\omega) = \frac{1}{\sqrt{\lambda_k}} \int_{\mathcal{X}} y_0(\mathbf{x}, \omega) q_k(\mathbf{x}) d\mathbf{x} \quad \longrightarrow \quad \zeta_k^{(n)} = \frac{1}{\sqrt{\lambda_k}} \sum_{j=1}^{n_g} w_j y_0(\mathbf{x}_j, \omega^{(n)}) q_{k,j}, \quad n = 1, \dots, N. \quad (9)$$

At this point, one may opt to represent each ζ_k from its N samples using a Gaussian approximation and so turning the KLE into a Gaussian process, or use kernel density estimation to represent more general processes. Both options allow the generation of additional samples for ζ_k (and in turn samples of y_0 through Eq. (8), and of y through Eq. (1)), making them useful surrogate models for conducting UQ. However, the relationship between ω to ζ_k is lost: if one wants to generate a surrogate estimate for a particular input parameter value ω , it would be unclear what corresponding value of ζ_k to use in Eq. (8). This difficulty becomes especially important if one plans to combine different KLEs, such as in a multifidelity framework.

C. Representing ζ_k via Polynomial Chaos Expansions

We seek to explicitly create mappings from ω to ζ_k , thus preserving the connection between the KLE and the original physical sources of uncertainty. This entails representing the random variable $\zeta_k : \omega \rightarrow \mathbb{R}$, and one possibility is through a finitely-truncated PCE (detailed descriptions of PCEs can be found in [7, 28–31]):

$$\zeta_k(\omega) = \sum_{\beta \in \mathcal{I}} b_{k,\beta} \Psi_{\beta}(\xi_1(\omega), \dots, \xi_{n_s}(\omega)), \quad (10)$$

where $b_{k,\beta}$ are the expansion coefficients, $\beta = (\beta_1, \dots, \beta_{n_s})$, $\forall \beta_j \in \mathbb{N}_0$, is a multi-index, $\mathcal{I}$ is a finite index set, n_s is the number of stochastic degrees of freedom in the PCE, ξ_j are reference random variables of the PCE with standard distributions, and $\Psi_{\beta}(\xi_1, \dots, \xi_{n_s})$ are multivariate polynomials of the product form

$$\Psi_{\beta}(\xi_1(\omega), \dots, \xi_{n_s}(\omega)) = \prod_{i=1}^{n_s} \psi_{\beta_i}(\xi_i(\omega)), \quad (11)$$

with ψ_{β_i} denoting degree- β_i orthonormal polynomials corresponding to the choice of ξ_i .

We clarify a few more choices for our PCE implementation. First, we set n_s to equal the number of uncertain parameters in the model, and further designate each ξ_i to be associated with the i th uncertain input parameter; this is often a very convenient and natural choice [31]. Second, while different options of ξ_i and ψ_{β_i} are possible under the generalized Askey family [32], we employ Legendre expansions with uniform $\xi_i \sim \mathcal{U}(-1, 1)$ for our turbulent jet application since our uncertain input parameters are all endowed with uniform distributions. As a result, only linear transformations are needed to convert between the $\xi_i \sim \mathcal{U}(-1, 1)$ and the uniform uncertain physical parameters, which can be done very easily. Third, we will employ total-order expansions of polynomial degree p , and so $\mathcal{I} = \{\beta : \|\beta\|_1 \leq p\}$, yielding a total of $n_t = (n_s + p)! / (n_s! p!)$ terms in the expansion. For now we also use the same basis for all ζ_k for simplicity, but mixing different polynomial families and degrees is certainly possible.

For each random variable ζ_k , we solve for its PCE coefficients b_{β} in a non-intrusive regression manner using the N

samples of ζ_k obtained from Eq. (9):

$$\underbrace{\begin{bmatrix} \Psi_{\beta^{(1)}}(\xi^{(1)}) & \cdots & \Psi_{\beta^{(n_t)}}(\xi^{(1)}) \\ \vdots & & \vdots \\ \Psi_{\beta^{(1)}}(\xi^{(N)}) & \cdots & \Psi_{\beta^{(n_t)}}(\xi^{(N)}) \end{bmatrix}}_A \underbrace{\begin{bmatrix} b_{k,\beta^{(1)}} \\ \vdots \\ b_{k,\beta^{(n_t)}} \end{bmatrix}}_b = \underbrace{\begin{bmatrix} \zeta_k^{(1)} \\ \vdots \\ \zeta_k^{(N)} \end{bmatrix}}_c, \quad (12)$$

where $\xi^{(n)} = \xi(\omega^{(n)})$ is the n th regression point (here ξ without subscript represents the vector of all ξ_i), A is a regression matrix of size $N \times n_t$ with its columns corresponding to the basis vectors and rows to the regression sample points. Often the number of simulations N available is very small, it is helpful to employ regularization techniques such as the (unconstrained) least absolute shrinkage and selection operator (LASSO) (i.e., ℓ_1 -regularization)

$$b^* = \arg \min_b \|Ab - c\|_2^2 + \tau \|b\|_1, \quad (13)$$

and the Tikhonov regularization

$$b^* = \arg \min_b \|Ab - c\|_2^2 + \tau \|\Gamma b\|_2^2, \quad (14)$$

where $\tau \geq 0$ is a regularization parameter and can be selected via cross-validation, and Γ is a chosen weight matrix. An example of using LASSO for PCE with cross-validation can be found in [22].

D. Multifidelity Karhunen-Loève Expansions

If we have the ability to generate simulation data from a HF model and a LF model, then we can write

$$y_{\text{HF}}(\mathbf{x}, \omega) = y_{\text{LF}}(\mathbf{x}, \omega) + [y_{\text{HF}}(\mathbf{x}, \omega) - y_{\text{LF}}(\mathbf{x}, \omega)] \quad (15)$$

$$\begin{aligned} &\approx \tilde{y}_{\text{LF}}(\mathbf{x}, \omega) + \tilde{y}_{\Delta}(\mathbf{x}, \omega) \\ &= [\tilde{\mu}_{\text{LF}}(\mathbf{x}) + \tilde{y}_{0,\text{LF}}(\mathbf{x}, \omega)] + [\tilde{\mu}_{\Delta}(\mathbf{x}) + \tilde{y}_{0,\Delta}(\mathbf{x}, \omega)] \\ &= [\tilde{\mu}_{\text{LF}}(\mathbf{x}) + \tilde{\mu}_{\Delta}(\mathbf{x})] + [\tilde{y}_{0,\text{LF}}(\mathbf{x}, \omega) + \tilde{y}_{0,\Delta}(\mathbf{x}, \omega)] \\ &= \tilde{\mu}_{\text{MF}}(\mathbf{x}) + \tilde{y}_{0,\text{MF}}(\mathbf{x}, \omega) \\ &= \tilde{y}_{\text{MF}}(\mathbf{x}, \omega), \end{aligned} \quad (16)$$

where $\tilde{y}_{\Delta}(\mathbf{x}, \omega)$ approximates the difference term $[y_{\text{HF}}(\mathbf{x}, \omega) - y_{\text{LF}}(\mathbf{x}, \omega)]$, and $\tilde{y}_{\text{MF}}(\mathbf{x}, \omega)$ denotes the overall multifidelity approximation to the field QoI from the HF model. In particular, the training dataset used to construct $\tilde{y}_{0,\Delta}(\mathbf{x}, \omega)$ requires pairs of HF and LF model evaluations at the *same* ω values. Furthermore, these pairs also need to be available on a common grid of $\mathbf{x}$, which may require restriction, prolongation, or interpolation of solution variables if HF and LF use different meshes. If N_1 LF runs are used for building $\tilde{y}_{0,\text{LF}}$ and N_2 HF-and-LF run pairs are used for building $\tilde{y}_{0,\Delta}$, then the total computational cost is $(N_1 + N_2)$ LF simulations and N_2 HF simulations; in practice, we would like $N_2 \ll N_1$.

Substituting the truncated KLEs and the PCE representations for ζ_k , we obtain

$$\begin{aligned} \tilde{y}_{0,\text{MF}}(\mathbf{x}, \omega) &= \tilde{y}_{0,\text{LF}}(\mathbf{x}, \omega) + \tilde{y}_{0,\Delta}(\mathbf{x}, \omega) \\ &= \sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega) + \sum_{k'=1}^{k'_t} \sqrt{\lambda'_{k'}} q'_{k'}(\mathbf{x}) \zeta_{k'}(\omega) \\ &= \sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \left[\sum_{j=1}^{n_t} b_{k,\beta^{(j)}} \Psi_{\beta^{(j)}}(\xi(\omega)) \right] + \sum_{k'=1}^{k'_t} \sqrt{\lambda'_{k'}} q'_{k'}(\mathbf{x}) \left[\sum_{j'=1}^{n_t} b_{k',\beta^{(j')}} \Psi_{\beta^{(j')}}(\xi(\omega)) \right] \\ &= \sum_{j=1}^{n_t} \left[\sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) b_{k,\beta^{(j)}} \right] \Psi_{\beta^{(j)}}(\xi(\omega)) + \sum_{j'=1}^{n_t} \left[\sum_{k'=1}^{k'_t} \sqrt{\lambda'_{k'}} q'_{k'}(\mathbf{x}) b_{k',\beta^{(j')}} \right] \Psi_{\beta^{(j')}}(\xi(\omega)) \\ &= \sum_{j=1}^{n_t} \left[\sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) b_{k,\beta^{(j)}} + \sum_{k'=1}^{k'_t} \sqrt{\lambda'_{k'}} q'_{k'}(\mathbf{x}) b_{k',\beta^{(j')}} \right] \Psi_{\beta^{(j)}}(\xi(\omega)), \end{aligned} \quad (18)$$

where the second equality substituted the PCEs for ζ_k and $\zeta_{k'}$, third equality interchanged the finite summations, and the last equality combined the terms in the brackets since here we assume the same polynomial basis terms are used for all ζ_k and $\zeta_{k'}$. Once we have $\tilde{y}_{0, \text{MF}}$, we can obtain $\tilde{y}_{\text{MF}}$ by adding back $\tilde{\mu}_{\text{MF}}$ through Eq. (16).

We make a few remarks about Eq. (18). First, while we assume the use of a common PCE basis for all ζ_k and $\zeta_{k'}$ terms, a more general situation involving different families and degrees of PCEs can be straightforwardly accommodated by taking the union of all basis terms. Second, the eigenvalues and eigenvectors from $\tilde{y}_{0, \text{LF}}$ and $\tilde{y}_{0, \Delta}$ are not the same, and so the two summations inside the bracket in Eq. (18) cannot be further combined. Third, Eq. (18) is *not* in the form of a KLE and should not be interpreted as $\tilde{y}_{0, \text{HF}}$ from $y_{\text{HF}} \approx \tilde{\mu}_{\text{HF}} + \tilde{y}_{0, \text{HF}}$, though we will compare against a $\tilde{y}_{\text{HF}}$ built using only HF simulations in our numerical illustrations, in terms of both accuracy and computational cost.

IV. Numerical Results

A. EDDDES-SA Simulation of a Single-Stream Round Jet

We now study the turbulent flow exiting an isothermal round jet at a jet Mach number of $M_j = 0.9$. The Reynolds number based on the nozzle diameter of $D = 0.06223$ m is $\text{Re}_D = 1.1 \times 10^6$. To investigate the behavior of the jet in flight, we will explore a range of nozzle exit stagnation pressure conditions p_0 within $\pm 3\%$ around a nominal stagnation pressure $p_0^* = 10218.44$ Pa (i.e., p_0 can vary between $[0.97p_0^*, 1.03p_0^*]$), and inlet eddy viscosity ratios (EVR) μ_t/μ between $[1.0, 100.0]$. In particular, the variation of EVR is of high interest since it controls the variation of turbulence intensity level at the nozzle exit. The flight Mach number is fixed at $M_f = 0.2$. The jet in static conditions ($M_f = 0$) was previously studied in the framework of an EU Project Go4Hybrid (G4H) and simulated with similar hybrid RANS/LES methods employed in this work [12]. Three meshes used in that project, ranging from 1.6M cells (G1) to 8.8M cells (G3) were graciously shared by the G4H partners.

Our HF model is the EDDDES-SA solver in SU2 (Section III.A). We begin with a validation simulation at static conditions ($M_f = 0$, $M_j = 0.9$) and compare it against the experiment data from Bridges and Wernet [4]. A uniform inlet profile is imposed at the nozzle exit plane using the total pressure and temperature to match $M_j = 0.9$. The non-dimensional time step based on the nozzle diameter and jet axial velocity is 0.006. A total of 400 convective time units (CTU) are used to compute the turbulence statistics after an initial transient of 400 CTU. Figure 1 shows the fine-grained turbulence structure in the jet plume resolved by the EDDDES-SA solver, and Fig. 2 reveals the resolved scales in the jet shear layer on the meridian plane. The SLA subgrid scale model used by the EDDDES-SA solver is clearly capable for unlocking the Kelvin-Helmholtz instability in the early shear layer. The EDDDES prediction of the mean velocity and root-mean-square (RMS) of velocity fluctuations along the nozzle lip line (at $r = 0.5D$) are shown to be in excellent agreement with measurement even for G1 (see Fig. 3). In this paper, only the G1 mesh is used in our multifidelity study due to computational considerations.

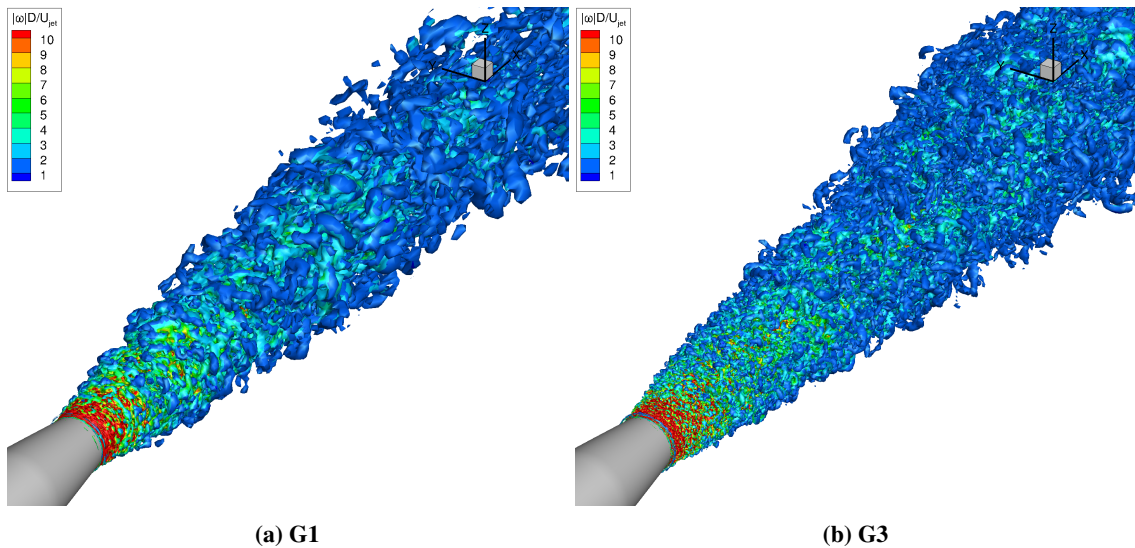


Fig. 1 Instantaneous Q criterion iso-surface colored by vorticity magnitude of G1 and G3

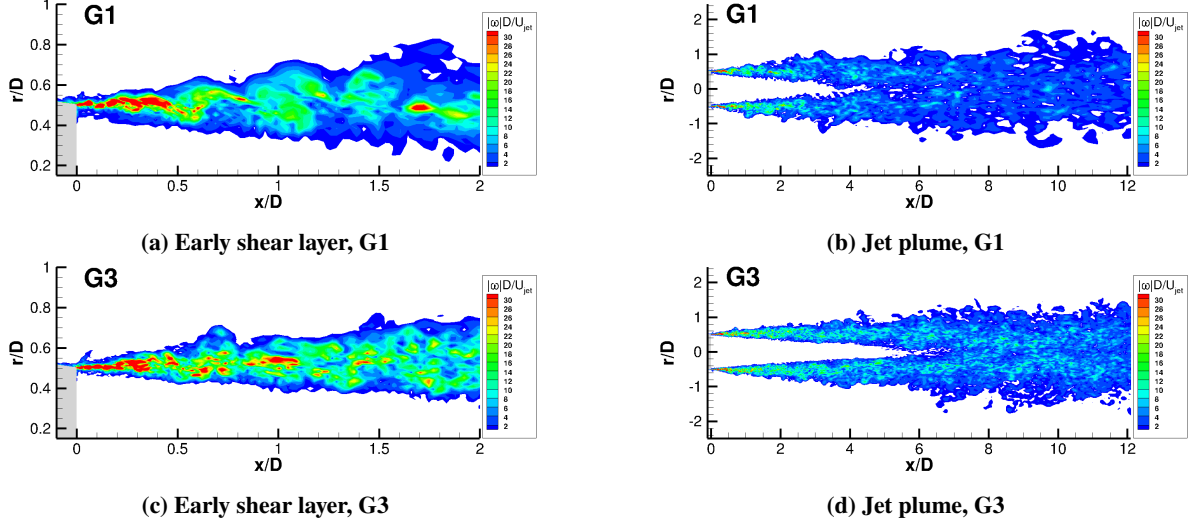


Fig. 2 Instantaneous vorticity magnitude contours in the early shear layer region ($0.0 < x/D < 2.0$) and the jet plume ($0.0 < x/D < 12.0$) on the meridian plane.

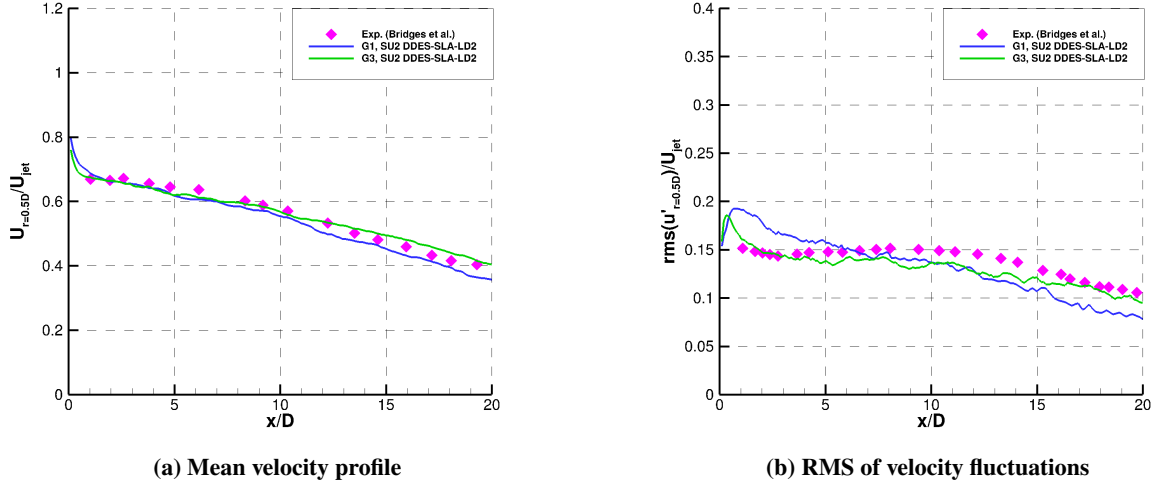


Fig. 3 Mean velocity and RMS of velocity fluctuations along the nozzle lip line ($r = 0.5D$).

We consider the following QoIs for the multifidelity framework outlined in Section III.D. The sound generation and emission of a turbulent jet is dominated by the influence of instability waves. The velocity fluctuations based on the normal Reynolds stress component $\overline{u'u'}$ along the nozzle lip line are strongly correlated with the instability waves in the jet shear layer and are known to be coherent over a long axial distance. Therefore it is considered as a QoI. Additionally, the shear Reynolds stress component $u'w'$ as well as the mean velocity along the lip line up to 20 diameters in the streamwise direction from the nozzle exit plane are also considered as QoIs. In total, we have three QoIs: mean velocity, $u'w'$, and $\overline{u'u'}$; they are all one-dimensional spatially varying field quantities since they are values along a single line in the x/D direction. These QoIs are extracted from the three-dimensional flow solutions using the open source data analysis and visualization software ParaView [33].

Our LF model is the steady RANS-SA solver, also implemented in SU2. The Reynolds stress components $\overline{u'u'}$ and $\overline{u'w'}$ required for the QoI's are computed using the Boussinesq assumption. The computational costs for the HF and LF simulations can be found in Table 1. Each HF run is about 260 times more expensive than a corresponding LF run.

Table 1 Comparisons of computational costs for HF and LF simulations, and total runs available for this study.

	CPU hours per run (approx.)	Number of runs
High-fidelity (EDDES-SA)	1.248×10^4	22
Low-fidelity (RANS-SA)	48	200

B. Multifidelity Karhunen-Loève Expansions and Uncertainty Quantification

To carry out the UQ analysis, we use data obtained from the LF (RANS-SA) and HF (EDDES-SA) turbulent round jet simulations to construct the multifidelity KLE following the method described in Sections III.B to III.D. The input parameters treated as uncertain are the stagnation pressure and the EVR. In this implementation, the EVR (which varies between $[1.0, 100.0]$) is first transformed to the logarithm of the modified eddy viscosity ratio (MEVR, $\tilde{\nu}/\nu$) [34] which now varies between $[1.53, 4.6]$. Uniform distributions are then prescribed for both parameters: $p_0 \sim \mathcal{U}(0.97p_0^*, 1.03p_0^*)$ where $p_0^* = 10218.44$ Pa, and $\log(\tilde{\nu}/\nu) \sim \mathcal{U}(1.53, 4.6)$ (i.e., $\tilde{\nu}/\nu$ has a log-uniform distribution). The input parameter distributions are summarized Table 2. Sample QoI profiles from the HF and LF simulations following these distributions are shown in Fig. 4. Overall, the LF mean velocity and $\overline{u'w'}$ follow similar trends as their HF counterparts but do not capture higher frequency oscillations. In contrast, the LF $\overline{u'u'}$ curves appear to have drastically different magnitudes and trend compared to the HF profiles (note their very different y-axis scales).

Table 2 Distributions for the uncertain inputs. The nominal stagnation pressure is $p_0^* = 10218.44$ Pa.

Parameter	Values	Distribution
Stagnation pressure, p_0	$[0.97p_0^*, 1.03p_0^*]$	Uniform
log-MEVR, $\log(\tilde{\nu}/\nu)$	$[1.53, 4.6]$	Uniform

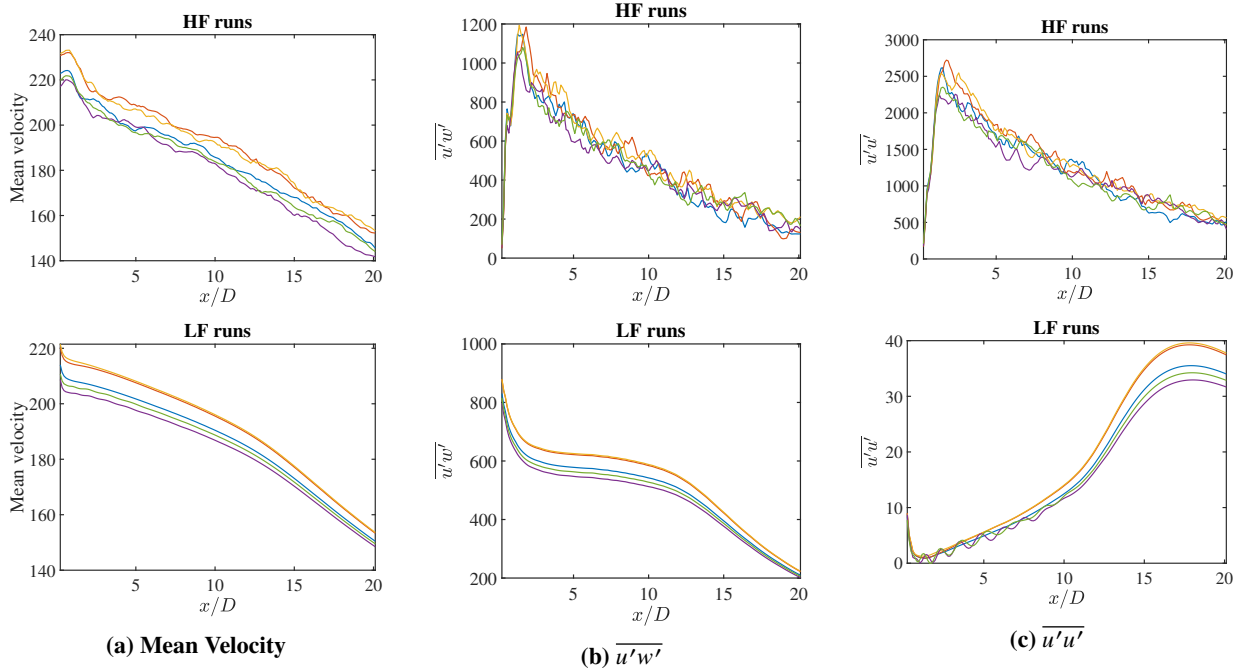


Fig. 4 QoIs from sample runs of HF (EDDES-SA) and LF (RANS-SA) models using the G1 mesh, extracted along the nozzle lip line ($r = 0.5D$).

For the multifidelity study in this paper, we created a database of 22 HF runs and 200 LF runs due to their high computational costs (Table 1), and 5 of these HF runs are designated as test points and untouched in the training process. While this relatively small database serves as an initial proof-of-concept, we plan to grow it as more simulations become

available in the future. With QoIs extracted from all simulations in this database, we use them to construct KLEs following the procedure outlined in Section III. We focus the comparisons among the following three surrogate cases:

- $\tilde{y}_{\text{HF}} = \tilde{\mu}_{\text{HF}} + \tilde{y}_{0,\text{HF}}$, using 17 HF runs only;
- $\tilde{y}_{\text{LF}} = \tilde{\mu}_{\text{LF}} + \tilde{y}_{0,\text{LF}}$, using 200 LF runs only; and
- $\tilde{y}_{\text{MF}}$ (Eq. (17)), consisting of a $\tilde{y}_{\text{LF}}$ using 200 LF runs, plus a $\tilde{y}_{\Delta}$ using 10 HF-and-LF run pairs.

For $\tilde{y}_{\text{MF}}$, we purposefully choose to use fewer HF runs (10) than in $\tilde{y}_{\text{HF}}$ (17), in order to illustrate that the advantages in $\tilde{y}_{\text{MF}}$ are attributed to the multifidelity construct (i.e., while having a HF saving), and not simply due to an addition of LF runs relative to $\tilde{y}_{\text{HF}}$.

Across all these surrogate cases, we then need to construct three distinct KLEs for each QoI: $\tilde{y}_{0,\text{HF}}$, $\tilde{y}_{0,\text{LF}}$, and $\tilde{y}_{0,\Delta}$. The eigenvalue spectra for these KLEs are shown in Fig. 5 (only the first 20 eigenvalues are shown for $\tilde{y}_{\text{LF}}$). We truncate all KLEs such that $\lambda_k/\lambda_1 \leq 0.01$, with truncation locations marked as red diamonds in the figures. They correspond to retaining 11, 15, 15 modes for the three QoIs in $\tilde{y}_{0,\text{HF}}$; 1, 1, 7 in $\tilde{y}_{0,\text{LF}}$; and 8, 9, 9 in $\tilde{y}_{0,\Delta}$. The spectral decay rates for $\tilde{y}_{\text{HF}}$ appear slower compared to $\tilde{y}_{\text{LF}}$, which is consistent with our intuition that a HF simulation (EDDES-SA) carries higher frequency information than LF (RANS-SA). It is difficult to comment on the $\tilde{y}_{0,\Delta}$ eigenvalue decay for the time being since we only have ten modes (training points) total. Sample eigenvectors (modes) from the HF KLE for $\overline{u'w'}$ are also shown in Fig. 6. Once the eigenvalues and eigenvectors are available, ζ_k samples can be calculated for each retained mode. The samples were checked to have mean very close to zero, variances to unity, and uncorrelated between different ζ_k ; these are all consistent with the Karhunen-Loève theorem. Using these samples, a PCE is created to explicitly map from ω to each ζ_k . Total-order polynomials with $p = 5$ are employed. LASSO is used for underdetermined settings while Tikhonov regularization is used for overdetermined settings, with regularization parameters selected from 20-fold (or leave-one-out if fewer than 20 points available) cross-validation.

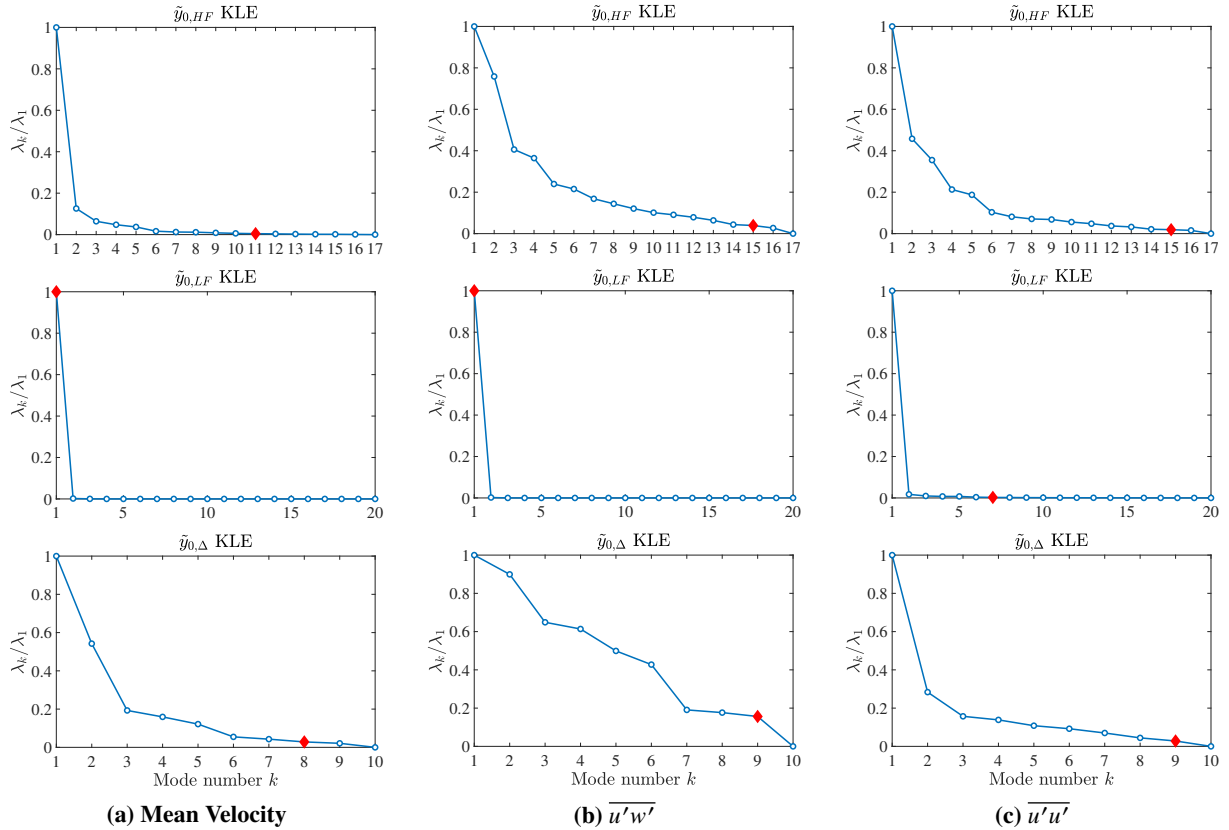


Fig. 5 Eigenvalue spectra for the three QoIs in $\tilde{y}_{0,\text{HF}}$, $\tilde{y}_{0,\text{LF}}$, and $\tilde{y}_{0,\Delta}$. The KLE truncations are made at the red diamonds.

At this point, all KLEs are complete. We can then use them as surrogate models in the following manner. At a given input condition $\omega = (p_0, \log(\tilde{\nu}/\nu))$ where we wish to make a prediction of the QoIs, we first transform ω into

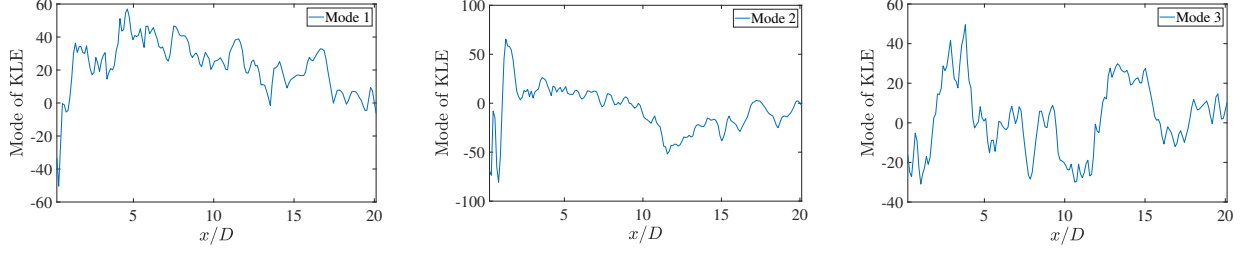


Fig. 6 The first three eigenvectors (modes) for $\overline{u'w'}$ in $\tilde{y}_{0,HF}$.

$\xi = (\xi_1, \xi_2)$ via a linear transformation since ω and ξ both follow uniform distributions but just with different bounds:

$$\xi_1 = \frac{2(p_0 - p_0^*)}{0.06p_0^*} \quad (19)$$

$$\xi_2 = \frac{2(\log(\tilde{\nu}/\nu) - \frac{1}{2}(1.53 + 4.6))}{(4.6 - 1.53)}. \quad (20)$$

The ξ values are then substituted into the PCEs to obtain ζ_k 's via Eq. (10), and ζ_k 's are carried into the KLE via Eq. (18) and Eq. (16) to arrive at a prediction of the QoI.

We have EDDDES-SA simulations at the five designated test points where we can compare the predictive capability of the three surrogate cases. First, we provide detailed results on two of the test points that exemplify the two types of behavior observed in this prediction analysis:

- Test point 1: $p_0 = 10154.43$, $\tilde{\nu}/\nu = 23.40$ ($\log(\tilde{\nu}/\nu) = 3.15$),
- Test point 2: $p_0 = 10498.71$, $\tilde{\nu}/\nu = 92.91$ ($\log(\tilde{\nu}/\nu) = 4.53$).

The QoIs for test point 1 are presented in Fig. 7, while for test point 2 in Fig. 8 (the second row of Fig. 8 is a zoomed in version of the first row). In order to meaningfully evaluate the behavior of the surrogates, we compare all predictions against the HF EDDDES-SA simulation y_{HF} at these two test points, also labeled as “Truth” in the plots. For Fig. 7, both $\tilde{y}_{MF}$ and $\tilde{y}_{HF}$ predictions capture the EDDDES-SA trend very well, even if they do not replicate all oscillations exhibited in the EDDDES-SA solution. The $\tilde{y}_{LF}$ captures mean velocity and $\overline{u'w'}$ well, but does poorly for $\overline{u'u'}$. However, with just 10 HF-and-LF pairs for $\tilde{y}_{MF}$, a reasonable approximation can still be recovered. This is an example of the multifidelity KLE overcoming the limitations of an inaccurate LF KLE. For the second test point in Fig. 8, $\tilde{y}_{HF}$ deviates significantly from the EDDDES-SA simulation. The $\tilde{y}_{MF}$, in contrast, is also able to produce a prediction that is much closer to the EDDDES-SA solution. This is an example of the multifidelity KLE overcoming the limitations of an inaccurate HF KLE. We provide further insight to explain this behavior below.

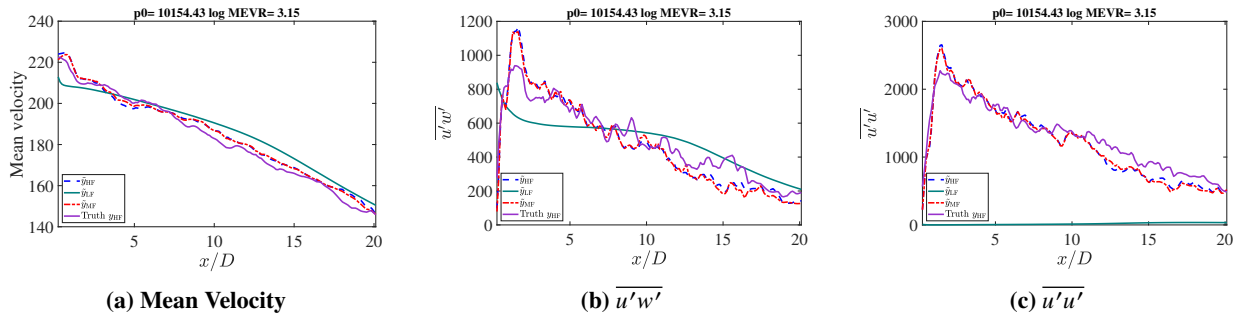


Fig. 7 Comparison of $\tilde{y}_{HF}$, $\tilde{y}_{LF}$, $\tilde{y}_{MF}$, together with the high fidelity EDDDES-SA simulation y_{HF} at test point 1.

We generate predictions at 900 additional test points, but we do not have the EDDDES-SA simulations y_{HF} at these new test points and therefore cannot compute the prediction errors. Instead, we analyze the relative *difference* between two surrogates ($\tilde{y}_{HF}$ and $\tilde{y}_{MF}$) defined as follows:

$$d(\omega) = \frac{\|\tilde{y}_{HF}(\mathbf{x}, \omega) - \tilde{y}_{MF}(\mathbf{x}, \omega)\|_2}{\|\tilde{y}_{HF}(\mathbf{x}, \omega)\|_2}. \quad (21)$$

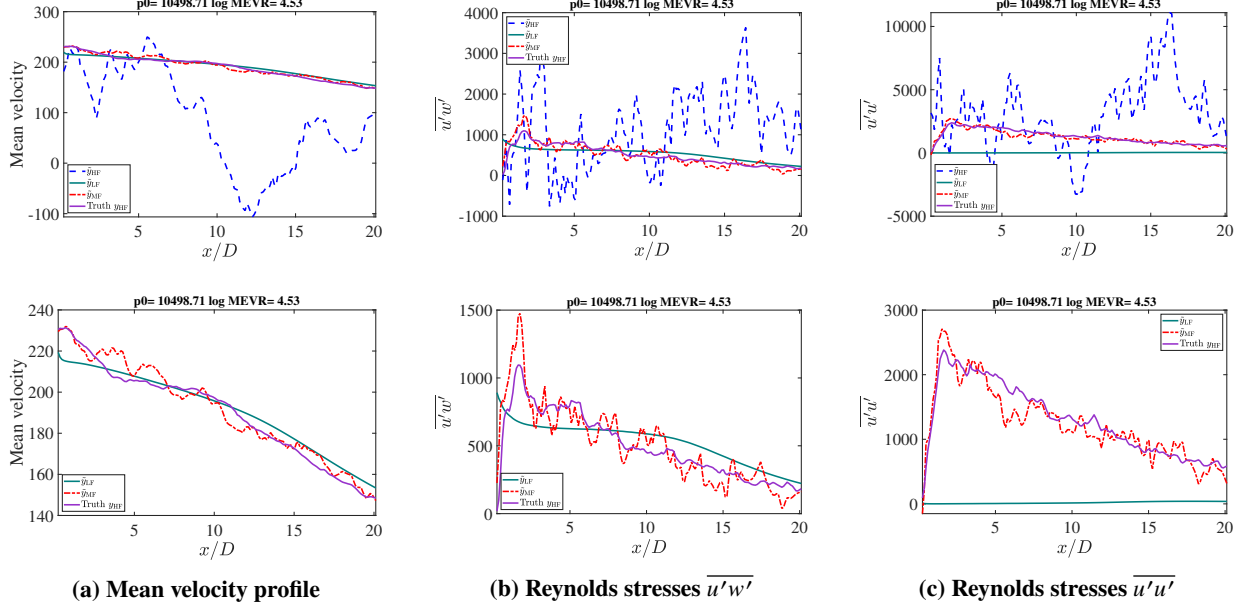


Fig. 8 Comparison of $\tilde{y}_{HF}$, $\tilde{y}_{LF}$, $\tilde{y}_{MF}$, together with the high fidelity EDDDES-SA simulation y_{HF} at test point 2. The bottom row shows a zoomed-in view for clarity.

The relative difference for all 900 test points are summarized in the histograms of Fig. 10. For the mean velocity QoI, a greater number of runs are observed to have low relative difference (ranging between 0 and 0.2), whereas the portion of low-difference predictions is smaller for the other two QoIs. The relative difference on QoI $\overline{u'u'}$ for all 900 test points is shown in Fig. 10, where the coloring of the points reflects the value of $\log(d)$, and the magenta diamond symbols indicate the locations of EDDDES-SA simulations used as training points for $\tilde{y}_{HF}$ (which is a superset of EDDDES-SA training points for $\tilde{y}_{MF}$). As expected, the relative difference increases at locations further away from the HF training points. Due to the very few HF simulations available, the region of high relative difference is large. However, we emphasize that a large d only indicates a disagreement between $\tilde{y}_{HF}$ and $\tilde{y}_{MF}$, but we do not know which one is more accurate. To explore further, we identify all five test points (including test points 1 and 2 highlighted earlier) for which we do have the HF EDDDES-SA simulation y_{HF} ; these points are marked by cyan triangles, and their $\overline{u'u'}$ QoI predictions are presented on the sides of Fig. 10. In particular, test point 1 shows that, while in good agreement, both $\tilde{y}_{HF}$ and $\tilde{y}_{MF}$ approximate the HF EDDDES-SA simulation y_{HF} well. Test point 2 indicates that while in disagreement, it is $\tilde{y}_{MF}$ that approximates y_{HF} well while $\tilde{y}_{HF}$ has orders-of-magnitude higher errors. The other three test points follow the same behavior. These observations suggest that the multifidelity KLE surrogate remains robust even when used at input far from the available HF simulations, due to the help of a larger set of LF simulations. The single fidelity HF KLE, in comparison, does not enjoy this property.

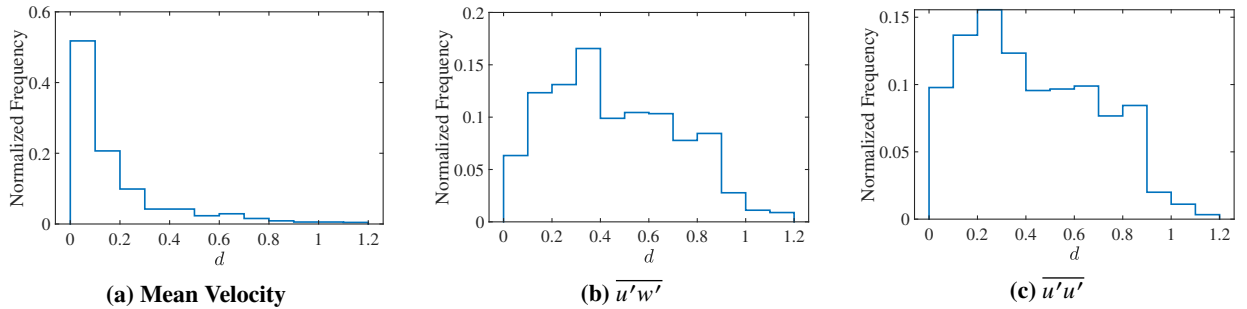


Fig. 9 Normalized histograms of relative difference for all 900 test points.

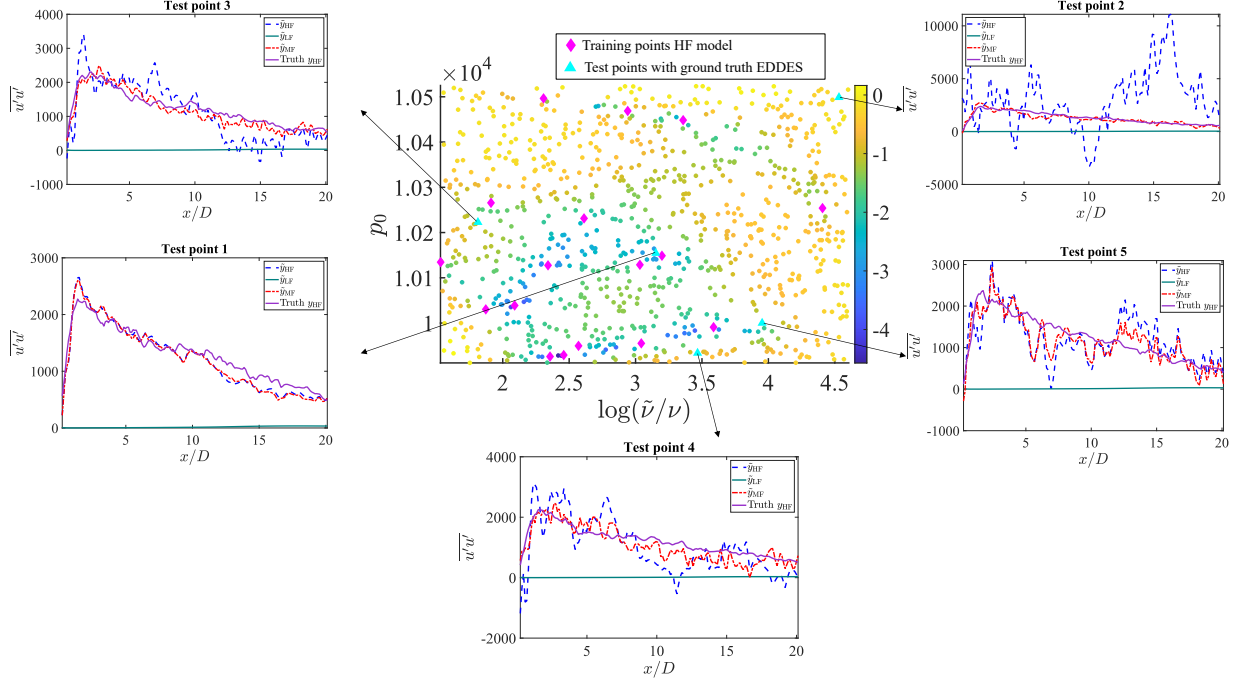


Fig. 10 Scatter plot with points whose color gradient reflects the log relative difference for QoI $\overline{u'u'}$ at different test points (i.e., $\log(d)$). The magenta diamond symbols indicate the locations of EDDES-SA simulations used as training points for $\tilde{y}_{\text{HF}}$. The cyan triangles indicate the five test points where their EDDES-SA simulations are available, and with their $\overline{u'u'}$ predictions presented on the sides.

Lasly, we conclude this section by presenting the UQ results obtained with the multifidelity KLE surrogate. The computational cost for evaluating the surrogate is negligible compared to either the HF or LF model. Figure 11 plots the three QoIs' prediction mean ± 2 standard deviations from evaluating the multifidelity KLE surrogate at 900 Monte Carlo samples. In this case, under the uncertainty presented in Table 2, we observe for mean velocity to have a heightened predictive uncertainty around 12 and 19 diameters down the lip line, while for $u'w'$ and $\overline{u'u'}$ they have relatively low uncertainty close to the nozzle exit while the two QoIs rise sharply.

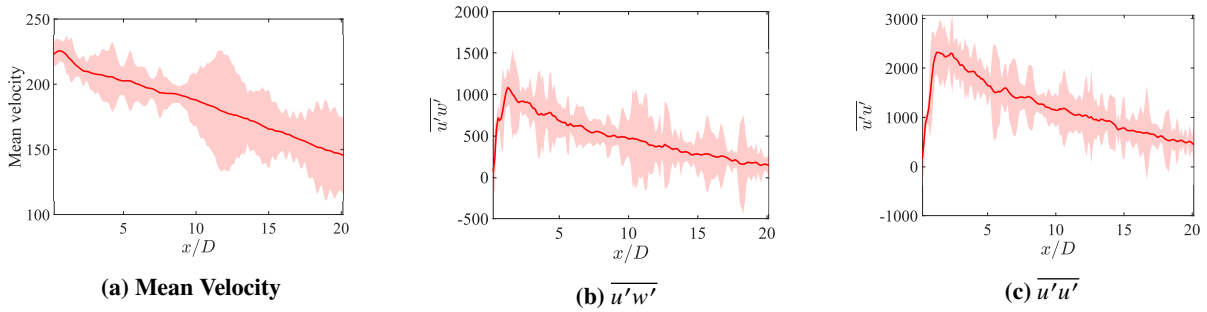


Fig. 11 QoI prediction mean ± 2 standard deviations from evaluating the multifidelity KLE surrogate at 900 Monte Carlo samples .

V. Conclusions

Understanding the behavior of turbulent jets under variable environment and uncertain conditions is critical for predicting and mitigating aircraft jet noise. An important step is to systematically investigate the impact of different sources of uncertainty and variability on the predicted QoIs. This can be achieved through performing UQ with

numerical simulations of the turbulent jets. However, jet noise requires high-fidelity eddy-resolving simulations where each is highly expensive, and the total computational requirement becomes prohibitive when many flow solves are needed such as in a Monte Carlo UQ analysis.

We approached this challenge by building surrogate models. In particular, we built KLE surrogate models for field QoIs in a multifidelity manner, by combining simulation data from both HF and LF models and taking advantage of their accuracy-versus-cost tradeoffs. We justified the combining procedure by representing the KLE random variables with PCEs, thus allowing an explicit association of the randomness in different KLEs to the same physical sources of uncertainty.

The new multifidelity KLE was demonstrated with simulations of a turbulent round jet, where we employed data generated from HF EDDES-SA and LF RANS-SA models. Under uncertainty from the nozzle exit stagnation pressure and inlet eddy viscosity ratio, we built KLEs for several one-dimensional spatially varying field QoIs: mean velocity, and the shear and normal Reynolds stress components. We compared at different test points the multifidelity KLE against single-fidelity KLEs built using only HF and only LF data, and illustrated the advantages of the multifidelity construct in terms of computational savings as well as robustness against extrapolation. In particular, we found that in situations where a HF KLE performs poorly, a multifidelity KLE can achieve more accurate predictions despite being constructed with fewer HF simulations than those used for the HF KLE. With the multifidelity KLE surrogate, we were able to conduct UQ for all QoIs inexpensively.

There are limitations to our work. We have not conducted a systematic analysis of the multifidelity KLE behavior under different combinations of HF and LF runs, which would be crucial for forming a model management system in guiding how many and in what proportions different runs are needed. Indeed, the observations in the turbulent round jet study are made with a relatively small dataset. While this reflects real life situations where expensive simulations are difficult to obtain and where a multifidelity approach is even more valuable, it would still be important to understand how the multifidelity behavior changes with larger or different datasets, and if the observed advantages persist. These are all interesting directions for continued future investigation.

Acknowledgments

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